Shubhit Sardana

Curriculum Vitae

Department of Physics
Indian Institute of Science Education and Research
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Inspire HEP *My Webpage* *Github*
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Education

2021–present **Integrated Bachelor & Master of Science**, *Indian Institute of Science Education and Research (IISER) - Bhopal, Madhya Pradesh - 462066, India.*
Cumulative Performance Index (CPI): 8.40 (from a maximum of 10)

Publications

Total Citations: 29, Published: 2, Pre-print: 2, In-prep: 2, My ADS Library

In Order of Relevance

Oct 2025 **[1]** Goncharov, B., **Sardana, S.**, Sesana, A. et al. Reading signatures of supermassive binary black holes in pulsar timing array observations. Nat Commun 16, 9692 (2025). https://doi.org/10.1038/s41467-025-65450-3, Published: **Nature Communications – 2nd Author**

Feb 2025 **[2]** Goncharov, B., & **Sardana, S.** (2025). Ensemble noise properties of the European Pulsar Timing Array. Monthly Notices of the Royal Astronomical Society, 537(4), 3470–3479. https://doi.org/10.1093/mnras/staf190, Published: **MNRAS – 2nd Author**

Dec 2025 **[3]** Nobleson, K., Dwivedi, C., Desai, S., et al. (2025), The Indian Pulsar Timing Array Data Release 2: II. Customised Single-Pulsar Noise Analysis and Noise Budget, https://arxiv.org/abs/2512.20455, Submitted: **Journal of High Energy Astrophysics – InPTA Collaboration**

Aug 2025 **[4]** Ojha, A., **Sardana, S.**, & Ghosh, A. (2025). Statistical Physics from Quantum Envariance Principles. arXiv [Quant-Ph]. Retrieved from http://arxiv.org/abs/2510.25253, Submitted: **Physical Review A – 2nd Author**

Research Experiences

June 2025 – present **Master's Thesis, Indian Institute of Science Education and Research**, *Bhopal*, India.
Thesis Supervisor: Dr. Mayuresh Surnis, IISER Bhopal, India
Co-Supervisor: M. A. Krishnakumar, NCRA, TIFR, India
Topic: Using Interplanetary Scintillation for Dispersion Measure Corrections in PTA Datasets

- Applied interplanetary scintillation (IPS) techniques to quantify heliospheric electron density variations and associated dispersion measure (DM) fluctuations in signal from Virgo A.
- Developed mathematical models of heliospheric plasma and validated them using Murchison Widefield Array (MWA) archival data through calibration, cleaning, and imaging pipelines.
- Currently extending the analysis to Ooty Radio Telescope (ORT) archival observations to construct heliospheric electron density maps and compute corresponding DM corrections for PTAs.

August 2025 – present **Research Intern, Max Planck Institute for Gravitational Physics**, *Hannover*, Germany.
Supervisor: Dr. Boris Goncharov
Topic: The Targeted Siren Cosmology with Pulsar Timing Arrays

- Conducting precision cosmology using Pulsar Timing Arrays (PTAs) to constrain Hubble constant through targeted continuous gravitational wave (CGW) searches.
- Analyzing realistically simulated CPTA and MPTA datasets to search for CGW signals from known SMBHB with strong electromagnetic counterparts and detectable strain amplitudes.
- Performed CGW signal injection and Bayesian parameter estimation. Combining inferred luminosity distance with redshift measurements to obtain constraints on Hubble constant.

Feb 2025 – June 2025 **Research Collaboration, Indian Pulsar Timing Array (InPTA)**.
Noise Analysis Group, InPTA
Topic: Incorporating Solar Wind Gaussian Process in Pulsar Noise Modeling

- Incorporated solar wind effects into pulsar noise models for low solar elongation pulsars in the InPTA DR2 dataset.
- Developed data-analysis pipelines using a Gaussian-process framework to model solar wind–induced timing variations and validated the approach across multiple pulsars.
- Regularly presented progress and contributed to methodological refinements through collaborative discussions in weekly group meetings.

May 2024 – Aug 2024 **Research Intern, Max Planck Institute for Gravitational Physics**, *Hannover*, Germany.
Supervisor: Dr. Boris Goncharov (Senior Scientist)
Topic: Ensemble noise properties of the European Pulsar Timing Array

- Implemented a novel prior-reweighting methodology developed by Dr. Boris Goncharov to marginalize over the uncertainties in pulsar noise priors using the EPTA DR2 dataset.
- Performed Hierarchical Bayesian inference to infer GWB parameters, enabling a more robust and statistically consistent characterization of PTA noise properties.
- Demonstrated that incorporating updated prior distributions increases the bayesian evidence for the gravitational-wave background (GWB) by improving consistency between inferred GWB parameters and theoretical expectations while mitigating systematic measurement biases.
- This work resulted in two peer-reviewed research publications: *"Ensemble Noise Properties of the European Pulsar Timing Array"* (MNRAS) and *"Reading Signatures of Supermassive Binary Black Holes in Pulsar Timing Array Observations"* (Nature Communications).

Dec 2023 – Apr 2024 **Ludwig Maximilian University of Munich (LMU Munich)**, *Munich*, Germany.
Supervisor: Dr. Karan Molaverdikhani
Topic: Analysis of EVLA Archival Data for Exoplanet Host Star Radio Emissions

- Analyzed the TRAPPIST-1 system to search for potential star–planet interaction signatures in radio wavelengths using archival EVLA observations.
- Processed, calibrated, and imaged extensive radio datasets using CASA and standard VLA data-reduction pipelines, followed by detailed analysis from the resulting images.
- Found no definitive evidence for radio emission associated with star–planet interactions, likely due to sensitivity limitations of current radio telescopes.
- Identified next-generation facilities such as the ngVLA and SKA-Mid as promising instruments capable of achieving the sensitivity required for future detections of such interactions.

May 2023 – Dec 2023 **Research Intern, Indian Institute of Science Education and Research**, *Bhopal*, India.
Supervisor: Dr. Mayuresh Surnis
Topic: Pulsar Search and Analysis Pipeline Development for GMRT Data

- Built and optimized a PRESTO-based pulsar search pipeline for GMRT data, including filterbank generation, RFI mitigation (RFI Clean), and de-dispersion.
- Improved computational performance by replacing single-frequency shifting with a sub-band shifting scheme, reducing I/O overhead while preserving accuracy.
- Performed accelerated pulsar searches using `accelsearch` and validated pulsar candidates.
- Acquired hands-on experience in pulsar signal processing, pulse profile analysis, and end-to-end detection workflows.

Conference & Teleconference Presentations

5th March 2025 **Indian Pulsar timing Array (InPTA)-wide Teleconference**.
Topic: (Research Paper) Exploring the time variability of the solar wind using LOFAR pulsar data. Presented on the importance of solar wind in PTA. Mentioned the need for integrating these effects and presented the results from the script built by me to analyze the process.

17th Oct 2024 **International Pulsar Timing Array (Gravitational-wave Analysis) Teleconference**.
Topic: (Two Research Papers) Ensemble noise properties of pulsars in EPTA and implications. GWA is one of the several working groups in the IPTA. The presentation focused on the key findings from the work done under the supervision of Dr. Boris Goncharov. This work has also been published in the form of two research papers.

29th July 2024 **European Pulsar Timing Array (EPTA)-wide Teleconference**.
Topic: (Research Paper) Ensemble noise properties of pulsars in EPTA.
Presented in the EPTA-wide Telecon, a bi-weekly EPTA meeting. The presentation focused on the Ensemble noise properties of Pulsars, a research project that I did under the supervision of Dr. Boris Goncharov.

Technical Skills

Programming Languages	Python (High Proficiency), C/C++ (Intermediate)
Statistical Techniques	Bayesian inference, Monte Carlo Markov Chain (MCMC), Hamiltonian Monte Carlo (HMC) & basic Frequentist inference
ML tools	Normalizing flows (within Variational Inference), JAX framework, Discovery-Nanograv (GPU)
Astronomy Tools	Enterprise-Pulsar & enterprise-extensions, Astropy, PRESTO (PulsaR Exploration and Search TOolkit), tempo2, CASA and WSClean (Interferometric imaging & data processing) and more..
Other Tools & Softwares	Linux (Proficient), High Performance Computing (Proficient), Bash/Shell scripting (Intermediate), Mathematica, HTML5, CSS.

Related Courses

Astronomy and Astrophysics, Radio Interferometry (NRAO Curriculum – self studied), General Relativity, Cosmology, Nuclear and Particle Physics, Statistical Mechanics, Special Relativity, Electrodynamics, Non-Linear Dynamics and Chaos, Numerical Methods and Programming, Classical Mechanics, Advanced Quantum Mechanics, Atomic and Molecular Physics, Mathematical Methods - 1 & 2.

Workshops & Certifications

- Attended the two-day *Winter Conference on Condensed Matter Physics, Astrophysics & High Energy Physics (Seventh Edition, 2024)*, conducted online and organized by The University of Tennessee, Knoxville, USA.
- Participated in the *Sagan Summer Workshop 2023*, a five-day online workshop hosted by the NASA Exoplanet Science Institute, California Institute of Technology (Caltech), Pasadena, CA.
- Attended a two-week online workshop, *Overview of Space Science (START Program)*, organized by the Indian Space Research Organisation (ISRO).
- Completed the Coursera course *The Evolving Universe*, taught by Dr. S. George Djorgovski (California Institute of Technology), focusing on the evolution of the Universe from early epochs to large-scale structure formation.